



Material Safety Data Sheet

Revision Date: 2019-12-05

Moon Tech(R) D-80

1. PRODUCT AND COMPANY IDENTIFICATION

1.1 Name of substance/preparation

Commercial product name: D-80

Chemical name: Dibutyltindilaurate

1.2 Use of substance/preparation

Industrial, intermediate chemical.

1.3 Company information

Manufacturer/distributor: ANHUI MOON TECHNOLOGY CO.,LTD

ADD: Room 449, Building 3, Industrial Park, 1151 Yanan South Road, Bengbu City, Anhui Province, China

Postal code: 233000

Tel/Fax: +86-552-2079969

1.4 Revision date

2016.05.12

2. HAZARDS IDENTIFICATION

2.1 Emergency overview

Classification according to Regulation (EC) No 1272/2008 [EU-GHS/CLP]

Acute toxicity(Inhalation)	Category 1
Acute toxicity(Oral)	Category 3
Skin irritation	Category 2
Eye irritation	Category 2A
Reproductive toxicity	Category 2
Specific target organ toxicity	Category 2
Acute aquatic toxicity	Category 1

ENVIRONMENTAL HAZARDS

Acute aquatic hazard	Category 1
Long-term aquatic hazard	Category 1

Label elements

Pictogram



Signal word

Danger

Hazard statement(s)

H301	Toxic if swallowed
H315	Causes skin irritation.
H319	Causes serious eye irritation.
H330	Fatal if inhaled.



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H361	Suspected of damaging fertility or the unborn child.
H373	May cause damage to organs through prolonged or repeated exposure.
H400	Very toxic to aquatic life.
H401	Very toxic to aquatic life with long lasting effects.

Precautionary statement(s)

P260	Do not breathe dust/ fume/ gas/ mist/ vapours/ spray.
P264	Wash hands and face thoroughly after handling.
P270	Do not eat, drink or smoke when using this product.
P273	Avoid release to the environment.
P280	Wear respiratory protection.
	Wear protective gloves, protective clothing, face protection.
P301+P310	IF SWALLOWED: Immediately call a POISON CENTER or doctor. Rinse mouth.
P302+P352	IF ON SKIN: Wash with plenty of water. If skin irritation occurs: Get medical advice or attention. Take off contaminated clothing and wash it before reuse.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P321	
P310	Immediately call a POISON CENTER or doctor/ physician.

Supple Classification according to EU Directives 67/548/EEC or 1999/45/EC

Symbol(s)	T+ - Very toxic N - Dangerous for the environment
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R-phrases(s)	
R25	Toxic if swallowed
R26	Very toxic by inhalation

Risk Combination Phrases

R36/38	Irritating to eyes and skin
R48/22	Harmful: danger of serious damage to health by prolonged exposure if swallowed
R50/53	Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment

2.2 Other effects

Chronic: Exposure limits have been recommended for organotin compounds to minimize the potential for adverse effects on immune function and the CNS.

3. COMPOSITION/INFORMATION ON INGREDIENTS

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Moon Tech(R) D-80**3.1 Chemical characterization (substance)**

Ingredients Name	CAS No.	Chemical characteristics
Dibutyltindilaurate	77-58-7	Substance

3.2 Hazardous ingredients

Chemical name	CAS No.	Concentration
Dibutyltindilaurate	77-58-7	≥95.0%
lauric acid	143-07-7	≤5%

4. FIRST AID MEASURES**4.1 General information**

Consult a physician. Show this safety data sheet to the doctor in attendance. Move out of dangerous area.

4.2 After inhalation

Remove from exposure and move to fresh air immediately. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical aid if cough or other symptoms appear.

4.3 After contact with the skin

Remove/Take off immediately all contaminated clothing. Gently wash with plenty of soap and water. Get medical advice/attention.

4.4 After contact with the eyes

Rinse cautiously with water for 15 minutes. Remove contact lenses, if present and easy to do. Get medical advice/attention.

4.5 After swallowing

Immediately call a POISON CENTER or doctor/physician. Rinse mouth.

4.6 Advice for the physician

Treat symptomatically and supportively.

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated. Material is extremely destructive to tissue of the mucous membranes and upper respiratory tract, eyes, and skin., spasm, inflammation and edema of the larynx, spasm, inflammation and edema of the bronchi, pneumonitis, pulmonary edema, burning sensation, Cough, wheezing, laryngitis, Shortness of breath, Headache.

5. FIRE FIGHTING MEASURES**5.1 Flash point**

Flash point is about 113 °C.

5.2 Suitable extinguishing media

Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

5.3 Specific hazards during fire fighting

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Carbon oxides, Tin/tin oxides.

5.4 Special protective equipment for fire fighting

Determine the need to evacuate or isolate the area according to your local emergency plan. Use water spray to keep fire exposed containers cool.

5.5 Additional advice

Self-contained breathing apparatus and protective clothing should be worn in fighting large fires involving chemicals.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures: Use personal protective equipment. Keep people away from and upwind of spill/leak. Ensure adequate ventilation. Entry to non-involved personnel should be controlled around the leakage area by roping off, etc.

Environmental precautions: Be careful not to let it flow into rivers, etc., since adverse effects on the environment are concerned. Methods and materials for containment and cleaning up: Absorb spilled material in a suitable absorbent (e.g. rag, dry sand, earth, saw-dust). In case of large amount of spillage, contain a spill by bunding. Adhered or collected material should be promptly disposed of, in accordance with appropriate laws and regulations.

7. HANDLING AND STORAGE**7.1 Handling****Precautions for safe handling**

Avoid contact with skin and eyes. Avoid inhalation of vapour or mist.

Precautions against fire and explosion

Normal measures for preventive fire protection.

7.2 Storage

Store in a tightly closed container. Store in a cool, dry, well-ventilated area away from incompatible substances. Do not store in metal containers. Store protected from moisture. Store protected from light.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION**8.1 Exposure limits**

Component	CAS	Source	Value
Dibutyltindilaurate	77-58-7	ACGIH, TWA	0.1 mg/m ³
Dibutyltindilaurate	77-58-7	ACGIH, STEL	0.2mg/m ³
Dibutyltindilaurate	77-58-7	ACGIH, SKIN_DES	Can be absorbed through the skin.
Dibutyltindilaurate	77-58-7	OSHA Z1A, TWA	0.1 mg/m ³
Dibutyltindilaurate	77-58-7	OSHA Z1A, SKIN_FINAL	Can be absorbed through the skin.

8.2 Exposure in the work place limited and controlled**8.2.1 Exposure in the work place limited and controlled**

Respiratory Protection: Use respiratory protection unless adequate local exhaust ventilation is

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provided or exposure assessment demonstrates that exposures are within recommended exposure guidelines. IH personnel can assist in judging the adequacy of existing engineering controls.

Suitable respirator: Organic Vapor Type.

Eye Protection: Under normal conditions, wear chemical safety glasses.

Hand protection: Chemical protective gloves should be worn.

Skin protection: Wash at mealtime and end of shift. Skin contact must be avoided by using impervious protective clothing (glove, aprons, boots.) Use chemical protective gloves as a minimum and wash skin promptly upon any skin contact.

Hygiene Measures: Remove contaminated clothing immediately. Exercise good industrial hygiene practices. Wash after handling, especially before eating, drinking or smoking.

8.2.2 Exposure to the environment limited and controlled

Local Ventilation: Recommended

General Ventilation: Recommended

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state/form	: Liquid
Color	: Clear yellow
pH	: 6.5~7
Boiling point	: >204°C at 12mmHg
Melting point	: 22~24°C
Flash point	: 113°C
Refractive index n_D^{20}	: 1.471 (20°C)
Molecular weight	: 631.56g/mol
Ignition temperature	: No data available
Oxidizing properties	: No
Upper explosion limit	: No data available
Lower explosion limit	: No data available
Freezing point	: 8°C
Vapor pressure	: 0.2mmHg(160°C)
Vapor density	: No data available
Density/relative density	: 1.066g/cm ³ (25°C)
Water Solubility	: <0.1g/100mL (20°C)
Heat of combustion	: No data available
Viscosity(dynamic)	: No data available

10. STABILITY AND REACTIVITY

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Moon Tech(R) D-80**10.1 General information**

Stable at room temperature in closed containers under normal storage and handling conditions.

10.2 Chemical stability: Normal

10.3 Reactive

Conditions to avoid: Light, moisture, excess heat, sparks, strong acids, strong bases, strong oxidizing agents and source of ignition.

Hazardous Decomposition Products: Carbon monoxide, irritating and toxic fumes and gases, carbon dioxide, tin/tin oxides

Hazardous polymerization: Hazardous polymerization will not occur.

11. TOXICOLOGICAL INFORMATION

11.1 Route of Exposure: Inhalation, skin contact and accidental ingestion.

11.2 Signs and Symptoms of Overexposure:

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated. Material is extremely destructive to tissue of the mucous membranes and upper respiratory tract, eyes, and skin., spasm, inflammation and edema of the larynx, spasm, inflammation and edema of the bronchi, pneumonitis, pulmonary edema, burning sensation, Cough, wheezing, laryngitis, Shortness of breath, Headache.

11.3 Acute Toxicity:

Chemical Name	CAS No.	LD50 (Oral)	LD50 (Dermal)	LC50 (Inhalation)
Dibutyltindilaurate	77-58-7	175mg/kg (Mouse)	> 2,000 mg/kg (Rabbit)	0.15 mg/L/2h (Mouse)

Inhalation May be fatal if inhaled. Causes respiratory tract irritation.

Ingestion Harmful if swallowed.

Skin May be harmful if absorbed through skin. Causes skin irritation.

Eyes Causes serious eye irritation.

11.4 Chronic Toxicity

No data available.

11.5 Other Health Hazard Information:

RTECS: WH7000000

12. ECOLOGICAL INFORMATION**12.1 Aquatic and Terrestrial Ecotoxicity****Ecotoxicity Effects:**

Toxicity to fish LC50 - O.latipes - 17 mg/l - 48 h

Toxicity to Bacterial No data available.

12.2 Persistence and Degradability Water:

Organotin compounds released into the environment are not stable. Their degradation is accelerated by photochemical & biochemical processes. Degradation may take a few days or several weeks. The

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opposite process is biological methylation to mono-, di-, tri-, tetramethyltin. In estuaries, the concentration of methyltin increases with salinity. Expected to undergo bioconcentration.

12.3 Bioaccumulative Potential Bioaccumulation: No data available.

12.4 Mobility in Soil: No data available.

12.5 Additional Environmental Information: No data available.

13. DISPOSAL CONSIDERATIONS

Product disposal:	Dispose of in accordance with local regulations.
Packaging disposal:	Dispose of in accordance with local regulations.

14. TRANSPORT INFORMATION**14.1 Land transport regulations**

Class: 6.1
Packing group: II
UN-Number: 2788
Proper Shipping Name: ORGANOTIN COMPOUND LIQUID N.O.S
Technical name: Dibutyltindilaurate

14.2 Transport by sea

Class: 6.1
Packing group: II
UN-Number: 2788
Proper Shipping Name: ORGANOTIN COMPOUND LIQUID N.O.S
Technical name: Dibutyltindilaurate

14.3 Transport by air

Class: 6.1
Packing group: II
UN-Number: 2788
Proper Shipping Name: ORGANOTIN COMPOUND LIQUID N.O.S
Technical name: Dibutyltindilaurate

15. REGULATORY INFORMATION**15.1 National regulations**

National and local regulations must be observed. For information on labeling please refer to section of this document.

15.2 Chemical inventories:**CHINA**

Safe management ordinance of dangerous chemical product (State Council announces on 2011): Safe use and production, the storage of a dangerous chemical, transport, loading and unloading were prescribed. The list of a dangerous chemical(2010)/CN-No.: 61857

WGK (Water Danger/Protection)



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CAS# 77-58-7: 2

Canada - DSL/NDSL

Listed on Canada's DSL List.

Canada – WHMIS

WHMIS: Not available.

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations and the MSDS contains all of the information required by those regulations.

Canadian Ingredient Disclosure List

Listed on the Canadian Ingredient Disclosure List

16. OTHER INFORMATION

Prepared by Moon Tech New Material Inc.

(R) indicates Registered Trademark

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The above information relates only to the specific materials designated herein and may not be valid for such materials used in combination with any other materials or in any process or if the material is altered or processed, unless specified in the text.

-End of Material Safety Data Sheet-